Paper 6: Online Privacy and Informed Consent — The Dilemma of Information Asymmetry

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Overview

This paper studies whether internet users are actually capable of giving informed consent to cloud service providers' privacy policies. Rather than only asking people what they think about privacy (opinion-based), the authors designed a two-part survey that tests what users actually know about how their data is collected, stored, and used — treating privacy literacy like a measurable knowledge domain.

Methodology

- Two-part online survey distributed at a large U.S. Midwestern university (students, staff, faculty).
- Part 1 (Knowledge, n=455): 42 questions (28 multiple-choice, 14 true/false) across 5 sections — Cloud Computing, Online Security, Economics of the Internet, Educational Records (FERPA), and Legal Aspects of Privacy.
- Part 2 (Opinions & Behavior, n=756): assessed online behavior, personal privacy attitudes, and views on cloud service providers.
- Framework: informed consent requires 5 principles (Friedman, Felton & Millett, 2000) — disclosure, competence, comprehension, voluntariness, and agreement. This study focuses on comprehension and voluntariness, the two hardest to achieve.

Key Findings — Knowledge Gaps

- Mean overall knowledge score: 59.5% — middle 50% of scores fell between 52-70%.
- Over half of respondents did not know Twitter and Netflix are cloud-based services.
- ~25% could not correctly answer a basic question about what cloud computing is.
- 44% did not know free websites can profit by selling user data directly to marketing companies.
- Education level and (surprisingly) age both significantly predicted higher knowledge scores — younger users were not more knowledgeable.
- Weakest section overall: Legal Aspects of Privacy / FERPA (educational privacy law).

Key Findings — Opinions & Behavior

- 81% of respondents said they had submitted information online that they wished they hadn't had to — evidence of coercion/lack of voluntariness.
- Only 43% had ever refused to use a website because of its Privacy Policy or Terms of Service.
- 89% opposed online tracking without permission, yet most still use tracked services daily — a clear instance of the 'privacy paradox.'
- When told how cloud providers actually use uploaded content (e.g., searching content for ad targeting), disapproval rates were very high (68-94%), despite most having already agreed to such terms.

- 78% felt current U.S. privacy laws are too weak; 72% would accept fewer features in exchange for more control over their data at no extra cost.

Conclusion / Takeaway

The study provides strong evidence of an **information asymmetry** between users and cloud service providers: companies know exactly what they do with user data; users largely do not. Because users lack both the knowledge to comprehend privacy policies (comprehension) and any real alternative to "accept or don't use the service" (voluntariness), the consent they give cannot be considered fully informed. This helps explain the privacy paradox — the gap between people's stated concern for privacy and their actual data-sharing behavior may stem not from indifference, but from an inability to understand or meaningfully choose. The authors suggest possible remedies: simplified/layered privacy policies, standardized legal requirements, privacy "nutrition labels," or individualized Knowledge-based Privacy Plans (KIPPs).

How the Two Papers Connect

Paper 2 explains *why* informed consent fails at the human level — users lack the comprehension and real choice needed to consent meaningfully. Paper 1 shows *how* this failure is actively reinforced at the interface level — through dark patterns specifically engineered to exploit that same lack of comprehension and lack of real choice. Read together: one paper diagnoses the user's knowledge deficit, the other diagnoses the industry's design practices that profit from it.